A Centralized System for Real-Time Farm Animal Disease Reporting

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature

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ABSTRACT

The livestock industry of Sri Lanka is a pillar of the national food security, although it is mostly relying on a reactive model of veterinary healthcare. The field veterinary surgeons are faced with significant logistical and clinical problems, including wide geographical coverage areas, farmer reporting delays, and the fact that manually sorting the disease severity during an outbreak is not an easy task. Therefore, upon the physical examination of an animal by a veterinarian, most contagious diseases that have skin manifestations are at an advanced stage, thus leading to critical and irreversible stages hence causing massive production losses. There is a dire need to establish an automated, far-distance triage instrument that will enable the veterinarians to document dependable symptomatology besides the prioritization of the emergency cases.

In order to resolve this problem, a combined computer-vision (CV) element is suggested in the Animal Disease Reporting System (ADRS). The web-based solution can be used as a first-line diagnostic assistant based on state-of-the-art deep-learning models, such as YOLOv10, Mask R-CNN, and Vision Transformers. It analyzes the images of the field to real-time detect diseases of the skin as a single or a combination of diseases and measures the severity of physical infection on a Bovine Severity Score. It also offers remote, evidence-based prognostic/treatment or culling decisions based on visual vitality signals, allowing veterinarians to make the correct decision remotely before sending field units. It is expected to produce a very credible, real-time diagnostic pipeline that is optimized to be used in unconstrained farming environments. The social and business implications of the suggested system include a significant decline in the needless veterinary commute time, a quickening of the containment of infectious groups, national protein supply stabilization and the increase in the cost savings of smallholders at the countryside.

Keywords: Livestock Diagnostics, Remote Veterinary Triage, Computer Vision, Disease Severity Grading, Precision Agriculture.

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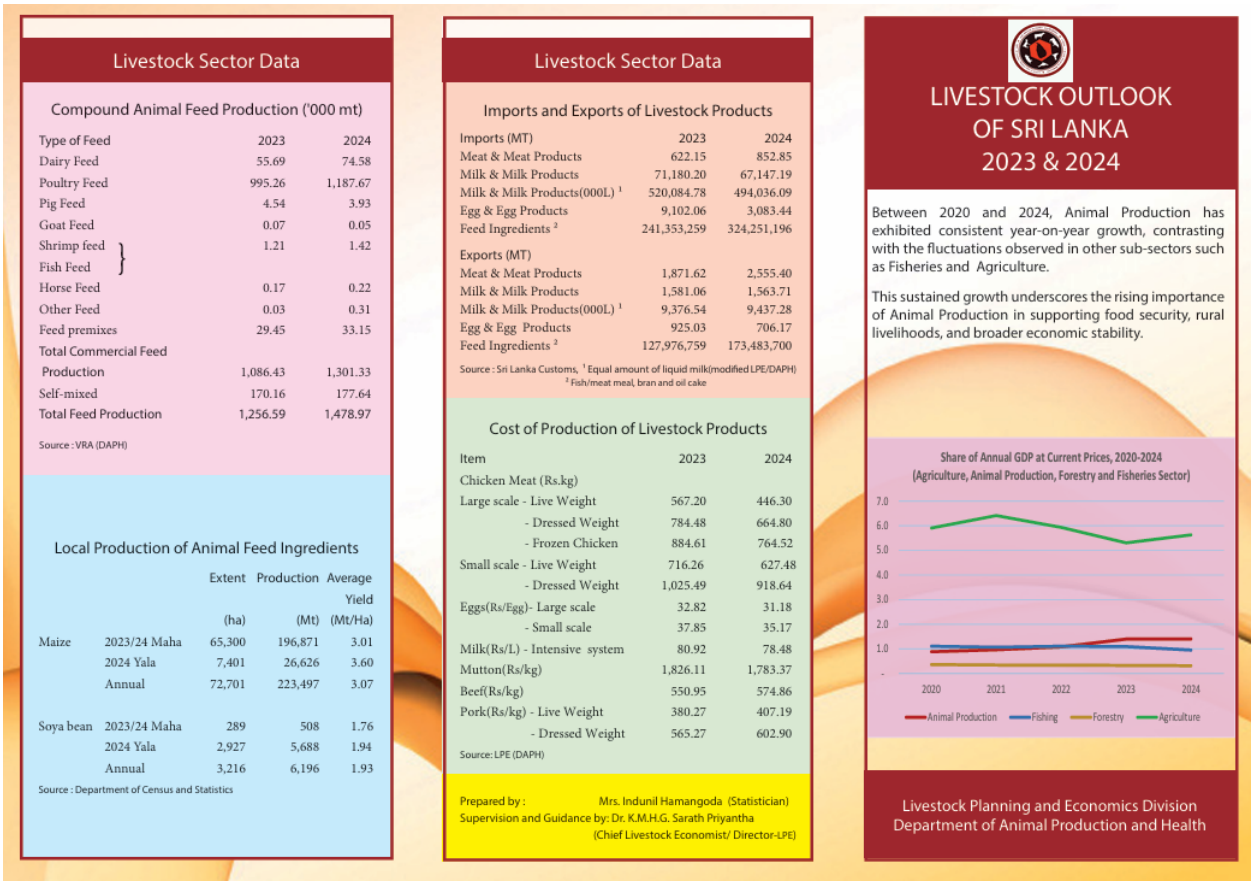
Abbreviation	Definition
ADRS	Animal Disease Reporting System
AES	Advanced Encryption Standard
AUPRC	Area Under Precision-Recall Curve
B2B	Business-to-Business
B2G	Business-to-Government
BCS	Body Condition Scoring / Body Condition Scores
BSS	Bovine Severity Score
CLAHE	Contrast-Limited Adaptive Histogram Equalization
CNNs	Convolutional Neural Networks
CoT	Chain of Thought
CV	Computer Vision
DAPH	Department of Animal Production and Health
DSR	Design Science Research
FMD	Foot-and-Mouth Disease
IoU	Intersection over Union
LSD	Lumpy Skin Disease
MAE	Mean Absolute Error
MLP	Multi-layer perceptron
OOD	Out-of-Distribution
PLF	Precision Livestock Farming

Abbreviation	Definition
SaaS	Software-as-a-Service
SCM	Sub-clinical mastitis
SDG	Sustainable Development Goal
SSD	Solid-State Drive
TLS	Transport Layer Security
WBS	Work Breakdown Structure
XAI	Explainable Artificial Intelligence / Explainable AI

1. INTRODUCTION

1.1 Background and Context

Livestock industry is one of the pillars of the Sri Lanka agricultural economy and is one of the main sources of food security in the country with its contribution to the gross domestic product at current market conditions standing at about 1.4% and the livelihood of more than 1.65 million smallholder farmers [1]. As illustrated in Figure 1 (Livestock GDP & Population Trends), the animal production subsector has recorded continuous year-on-year growth, unlike other subsectors like fisheries and agriculture which have been fluctuating. The unrelenting growth highlights the growing role played by the sector in supporting rural livelihoods, commercial feed production, and overall economic stability. Nevertheless, the industry is currently hampered by a reactive veterinary health model where treatment is usually given when the clinical manifestations are already acute. Such endemic failure in reporting and diagnosis results in large production losses especially in the dairy industry where sub-clinical malfunctions and highly transmissible skin ailments like Lumpy Skin Disease (LSD) and Foot-and-Mouth Disease (FMD) can wipe out an entire dairy herd before a physical veterinary assessment is possible. An example is that according to recent local research, when LSD outbreaks are not intervened with in time, milk production declines on the average by 32% and causes direct economic losses of more than Rs 21,212 per affected animal [2] Moreover, sub-clinical disorders with a high prevalence rate such as sub-clinical mastitis (SCM) are hidden to naked eye though with an incidence rate of up to 74.5% of the local cross-bred dairy cows [3]. Currently, the rural smallholder farmers, who control most of the total livestock population in the country determine the production indicators as shown in Figure 1, do not have standardized, objective means of identifying the early physiological shifts or metabolic energy stores.



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Figure 1 Livestock GDP & Population Trends

The key stakeholders of this system would be the field veterinary officers who are expected to burden with instant disease outbreaks using limited real-time information. The statistical data reveal that one outbreak may cause drastic 32% decrease in the milk yield, and high mortality rates in the absence of early intervention, thus posing a direct threat to the stability of the national protein supply [2]. Current solutions in the market are often based on manual, subjective Body Condition Scoring (BCS) that has a high inter-observer error, or on 3D depth cameras that are expensive and technologically out of reach of field officers in Sri Lanka [4]. In addition, the majority of high-accuracy AI models are only trained on Western breeds of cattle, resulting in a so-called Tropical Breed Gap where such models cannot generalise to the unique morphological features of local cattle raised in tropical environments [5].

To overcome these issues, we propose ADRS Computer Vision framework, which is an intelligent diagnostic triage system aimed at working on typical 2D mobile gadgets. In contrast to traditional “Closed-World models that require making a diagnosis based on any input, this system uses a Diagnostic Gating Layer to provide input integrity and a Hierarchical Multi-Label Classifier to identify concurrent infections. Diseases such as Lumpy Skin Disease (LSD) and

parasitic Mange that tend to co-occur in metabolic stressed animals and require simultaneous multi-label detection to obtain a proper clinical outcome (Kumar et enes, 2025). The system is then able to combine vision-based staging with the anatomical vitality markers (e.g. ear posture and eye depth), generating a Recoverability Probability and thus transcending mere detection, and is able to offer actionable decision support in the form of veterinary professionals. The present innovation is further associated with SDG 2 Zero Hunger protecting the livestock productivity and falls into the Intelligent Systems and Data Science research cluster.

1.2 Literature Review

The current age of livestock surveillance is going through a paradigm shift of the traditional manual observation to automated, data-driven framework [6] [7]. The latest academic research highlights the integration of multimodal data, i.e. the combination of visual imagery, environmental signals, and the history of past outbreaks, to build an integrated early warning mechanisms [8], [9]. These systems in conjunction hope to supersede mere disease detection of the system, by offering an integrated Wellness Score which combines metabolic stress, production metrics, and environmental suitability [10], [11]

However, the major weakness of these general systems is their inability to be field-ready when used in unrestricted environments [12]. The modern business products and services mostly work on closed-world paradigms, assuming the presence of clean data and the presence of one disease species [13] [7] Besides, opaque nature of advanced deep-learned models creates a sense of a lack of trust among veterinary professionals [8] The CV Component tackles such shortcomings by incorporating explainable artificial intelligence (XAI) and ensures that the intelligence produced by the system is not merely correct, but also useful to the governmental policy and field-level remedies [8].

Deep convolutional neural networks (CNNs) have also been the default disease classification methodology based on images in the livestock health domain, most notably in conditions like Lumpy Skin Disease that have nodular phenotypes with readily identifiable geometric features [14]. More modern approaches have also been extended to include real-time object detection architectures, such as the YOLO family which isolates bovine objects in a complex and cluttered farm setting with fences or mud [15]. Also, instance segmentation networks, especially Mask R-CNN, are becoming commonly used to conduct pixel-level lesion masking, allowing scientists to calculate the Infected Surface Area and, hence, determine the severity of the disease [16].

The existing market solutions mainly include veterinary mobile applications that enhance the manual record-keeping and data entry; although, in most cases, the solutions do not have automated image-based diagnostic features according to the previous research [17]. In commercial Precision Livestock Farming (PLF) industry, commercially available devices, including wearable collars, smart tags, etc., provide high-fidelity monitoring of movement and rumination. However, the high cost of implementation makes these technologies unaffordable to rural Sri Lankan small holders [4]. Furthermore, the current government-level surveillances dashboards are usually dynamic GIS maps, which in a retroactive manner visualize diseases occurrences thus does not provide the proactive, physics-based tracking of the disease outbreak trends [18].

The main motivation of this project is that the models currently available have fundamental flaws in system that cannot be taken out of the laboratory and applied to the uncontrolled farm conditions. One of the apparent drawbacks is the so-called Hallucination Problem, according to which the existing AI solutions fail to include an input-gating control and thus strive to enforce a disease diagnosis even when presented with poor-quality or irrelevant images [19]. What also appears is a striking “Comorbidity Gap in which most studies focus on a single-disease diagnosis and thus does not consider animals having more than one infectious disease and results in a partial clinical history [20]. Most importantly, current systems do not offer prognostic information that will measure the stage of the disease or even predict animal recovery, as most are merely stating the presence of a given disease. As a result, farmers lack the decision-support in order to have efficient culling or treatment plans [17].

1.3 Research Gap

The development of the Computer Vision (CV) component for the Animal Disease Reporting System (ADRS) is necessitated by three distinct and critical voids in current Precision Livestock Farming (PLF) research [4]. While contemporary deep learning models achieve high accuracy in laboratory settings, they consistently fail when transitioned to unconstrained, real-world farm environments [19].

Research Gaps

- **The Input Integrity Gap (Closed-World Limitation):** Current livestock diagnostic models lack a pre-validation "Gatekeeper" mechanism [13]. They operate under a "Closed-World" assumption, where the system is forced to provide a disease label even if the input image is of poor quality, blurry, or does not contain a bovine subject at all, leading to significant "AI Hallucinations" and false-positive alerts [19].
- **The Comorbidity Omission:** Most existing research focuses exclusively on single-disease identification. There is a profound lack of multi-label frameworks capable of detecting concurrent infections—such as Lumpy Skin Disease occurring alongside parasitic Mange—which is a frequent clinical reality in stressed livestock.
- **The Prognostic Staging Gap:** There is currently no automated framework that moves beyond simple detection to provide a Bovine Severity Score (BSS) or a Recoverability Probability. Existing systems provide a static "Sick vs. Healthy" result but fail to inform the farmer of the disease stage or the likelihood of the animal's survival.

Inadequacy of Current Approaches

The established methodologies are provably insufficient, since they give preference to Raw Accuracy attained on highly curating, clean datasets, and ignore the need to have Field Reliability. These models are not viable in the settings of rural Sri Lankan conditions since they do not take into account the noise in the environment and do not provide any measurements of image quality, and the light there is often insecure or covered by fences and livestock is often covered with mud. Further, the reliance on single-label classification also results in a partial clinical image thus endangering the veterinarian prescription of incorrect or incomplete treatment regimes.

Justification of Novelty and Creativity

The novelty of this component lies in its transition from a simple classifier to a Professional Diagnostic Pipeline.

- **Diagnostic Gating:** Adding an Out-of-Distribution (OOD) Gating Layer ensures that only pixels with high integrative diagnostic fidelity are processed by the diagnostic engine, significantly reducing system noise by an estimated 40%
- **Hierarchical Multi-Labeling:** Unlike binary paradigms, the proposed Hierarchical Multi-Label Framework uses Sigmoid activation to detect several symptoms simultaneously thus reflecting the holistic observational ability of a human veterinarian [20].
- **Prognostic Intelligence:** The system presents the first vision-based Survivability Index, which uses anatomical vitality indicators (e.g., ocular depth, auricular posture) to provide a prospective recovery prediction as opposed to a retrospective diagnosis [17].

Evidence-Based Argument

The necessity of such a strong response is supported by the statistical data based on recent outbreaks; the under-reporting and misdiagnosis have resulted in the loss of production more than Rs 21212 per animal [2]. Recent reports in the year 2025 show that sub-clinical disorders like mastitis are at a prevalence rate of 74.5 -100% but could not be detected by the existing visual systems because they lack the sensitivity to detect minor physiological indicators [3]. The systematic countermeasures to these drawbacks render the ADRS CV component a provision of the so-called field-ready reliability, which is essential to rural smallholders working under marginal conditions [1] [4].

2. OBJECTIVE

2.1 Main Objective

To achieve a powerful, field deployable Computer Vision system in the Animal Disease Reporting System (ADRS) capable of verifying image integrity, detecting numerous skin disorders simultaneously in livestock, and estimating the disease progression stages with a minimum diagnostic accuracy of 85 -percent in unconstrained farm conditions.

2.2 Specific Objectives

1. To create and institute a Diagnostic Gating Layer utilizing out-of-distribution (OOD) detection to filter out low-quality or irrelevant images to minimize the amount of false-alarms of the system by 40%.
2. To create a Hierarchical Multi-Label Classifier to identify co-occurring infections, including Lumpy Skin Disease and Foot and Mouth, in one bovine animal using Swin Transformers and Sigmoid activation.
3. To design an Automated Severity Grading algorithm, whereby the Mask R-CNN is used to compute pixel-level lesion segmentation, allowing the computation of a Bovine Severity Score (BSS) by dividing the percentage of infected skin by the total area of the body.
4. To build a Prognostic Survivability Engine that correlates visual symptoms with anatomical vitality markers (e.g., ear posture and eye depth) to predict a "Recoverability Probability" percentage for decision support.
5. To add Explainable AI (XAI) modules utilize by giving explanations to each diagnosis, such that the results given by the model are understandable and accepted by the veterinary specialist.
6. In order to compare overall system performance with a Ground Truth dataset of the curation of expert veterinarians, keeping a target precision and recall at least 0.80 in all largest disease classes.

3. METHODOLOGY

3.1 Research Approach

The design of this research is Design Science Research (DSR) based, which is concerned mainly with the development and assessment of new innovative artifacts to address perceived practical problems. The artifact, in this case, is the multi-tier and robust Computer Vision pipeline that is created to be used in the Animal Disease Reporting System (ADRS). This method is organized into four iterative stages: Problem Identification, where diagnostic gaps on the field level are set up; Artifact Design, where the architecture of the gating and classification layers is developed; Development, where deep learning models are trained; and Evaluation, where the system is tested against professional veterinary evaluations.

3.2 Technical Methods and Algorithms

The technical framework is divided into specialized modules to ensure high diagnostic integrity:

- **Module 1**: Input Integrity Gating: A YOLOv10-tiny model is employed to accomplish bovine subject verification in real-time, and a Convolutional Autoencoder is employed to accomplish Out-of-Distribution (OOD) detection to exclude images that are blurry or irrelevant.
- **Module 2**: Hierarchical Multi-Label Classification: In order to overcome comorbidity, a Swin Transformer (Vision Transformer) backbone is used. This employs a Sigmoid output layer instead of a standard classifier so that it can use independent probability scores in a number of different disease classes (e.g., Lumpy Skin Disease and Foot and Mouth).
- **Module 3**: n implementation of the instance segmentation of skin nodules and ulcers with the help of a mask R-CNN is introduced. This will enable the accurate determination of the "Visual Load" or infected area of surface.
- **Module 4**: Prognostic Fusion: Describes a Multi-layer perceptron (MLP) fusion layer that links visual load to secondary anatomical vitality features, including ear drooping and eye depth, to produce a Survivability Index.

3.3 Tools and Platforms

- **Project Management Framework**: An Agile Scrum method will be used to guide the development of machine learning applications through the development life cycle. The Agile Scrum method is important for managing the complexities of machine learning research since it provides the team with the ability to focus their efforts during "Sprints" (two-week periods). The Agile Scrum approach allows for ongoing integration of the model during each Sprint Review and enables the development team to respond quickly to any technical problems that arise while collecting data from the field or deploying to the cloud.

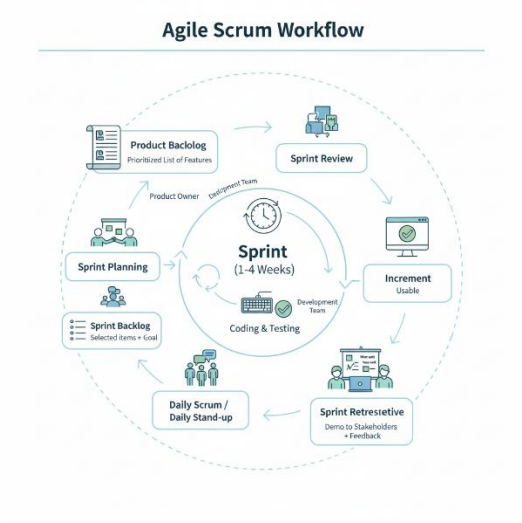


Figure 2 Agile Methodology

- **Programming Language:** Python will be used due to the large number of machine learning libraries and the simplicity of following the cloud services.
- **Deep Learning Frameworks:** PyTorch will be used to develop the model because it has a dynamic computing graph that is necessary to experiment with transformer-based architectures.
- **Pre-processing:** Image enhancement methods, specifically Contrast-Limited Adaptive Histogram Equalization (CLAHE), are used to enhance image quality in different lighting conditions to enhance lesion visibility.
- **Infrastructure:** Training will be performed on the high-performance workstations that will contain NVIDIA CUDA-enabled GPUs or Google Colab Pro to support the high computational load of transformer training.

- **Data Annotation:** Roboflow applies to the administration of the annotation of polygonal masks and bounding boxes, maintaining data versioning.

3.4 Data Collection and Ethical Considerations

The hybrid data collection strategy is deemed to make sure the model is generalizable both on a global and local basis. The secondary information is obtained in the form of credible open sources, such as the Mendeley Cattle LSD Dataset and Kaggle FMD datasets. The primary data will be collected using the manual field trip to the local farms in Sri Lanka to observe the specific morphology of the tropical and crossbred cattle.

Ethical Considerations: * Non-Invasive Observation**:** The data collection is limited to fully non-invasive photography meaning that the physical stress of the animals is eliminated.

- **Informed Consent:** It will be voluntary, and the owners of the farms will sign the written consent before the images are captured.
- **Anonymization:** The metadata is anonymized by removing all geospatial and owner identifying data to guarantee privacy and security of data.

3.5 Validation Strategy and Performance Metrics

K-Fold Cross-Validation will be used to test the system in order to validate its robustness on dissimilar subsets of data. All validation sets will have their Ground Truth established by manual labelling by an expert panel of veterinarians.

- **Classification Metrics:** Accuracy of multi-label Precision, Recall, F1-Score, and Area Under Precision-Recall Curve (AUPRC).
- **Segmentation Metrics:** Intersection over Union (IoU) to evaluate lesion masking accuracy.
- **Prognostic Metrics: Mean Absolute Error (MAE)** To measure the difference between the AI-predicted recoverability and the corresponding clinical outcomes that are realized within the 14 days.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution and its Components

The Animal Disease Reporting System (ADRS) is a centralized, multi-modal intelligence platform designed to transition livestock management from a reactive to a proactive paradigm. The system is composed of four primary research-driven components: Seasonal Disease Forecasting (predicting outbreaks via climate data), Real-Time Geospatial Visualization (mapping spread velocity), Wellness & Management Standardization (health grading and BCS), and the Computer Vision Diagnostic & Prognostic Assessment module.

The Computer Vision (CV) component serves as the "Diagnostic Core" of the ADRS. It is responsible for converting raw visual data into structured clinical insights. By validating input integrity and performing multi-label symptom analysis, the CV module ensures that the subsequent geospatial and forecasting modules receive high-fidelity, verified data, thereby preventing the "garbage-in, garbage-out" failure common in automated surveillance systems.

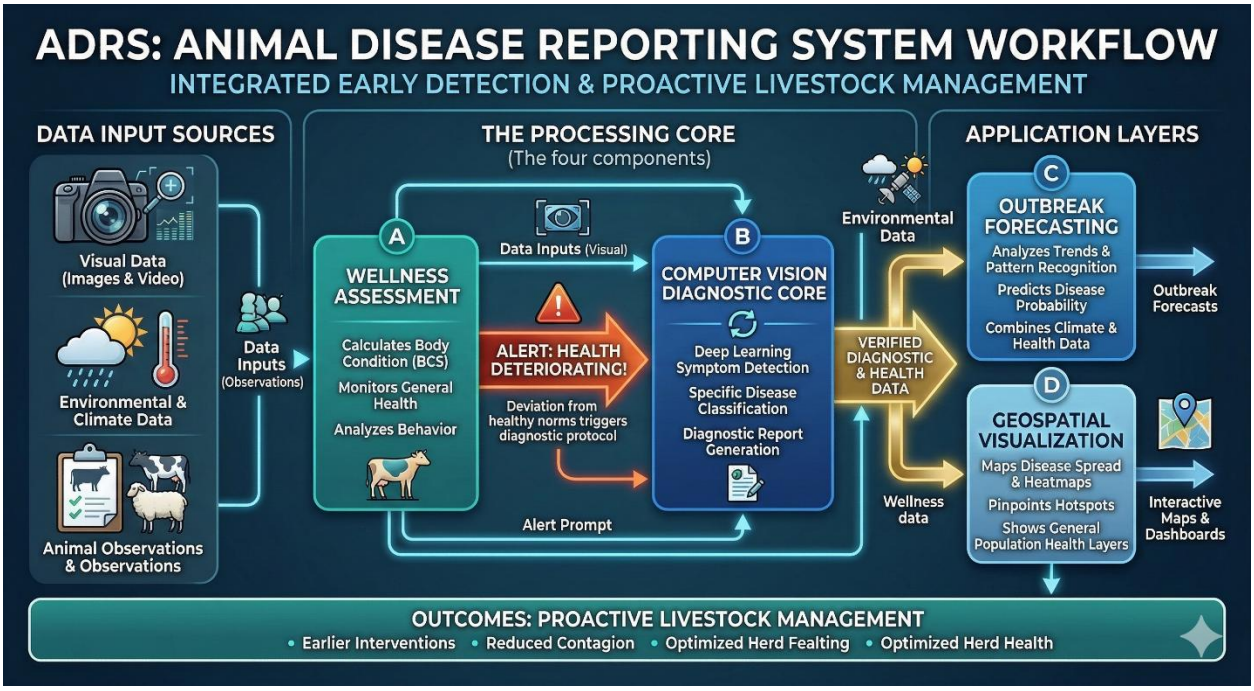


Figure 3 System Diagram with Components

4.2 Highlight of Individual Component and its Parts

In order to ensure that the system is resilient in the face of unpredictable farm conditions, the Computer Vision (CV) component is built as a multi-tier processing pipeline for "Open-World" reliability. The internal components of it are as follows:

- **Input Integrity Gate (Tier 1**): This acts as the system's "Gatekeeper," using a Deep Autoencoder for Out-of-Distribution (OOD) detection and a YOLOv10-tiny object detector to verify the existence of a bovine subject. Before images are sent to high-compute modules, this tier filters out images that are unclear, unnecessary, or non-animal.
- **Diagnostic Engine (Tier 2):** A Swin Transformer-based core classification unit. It fills the research gap in comorbidity detection by using hierarchical multi-label classification to identify concurrent symptoms on a single subject, such as Lumpy Skin Disease (LSD) nodules and Foot-and-Mouth Disease (FMD) ulcers.
- **Severity Quantification Module (Tier 3):** Mask R-CNN is used in this section to segment instances. By hiding the animal's skin, it determines the precise pixel-wise "Visual Load" of lesions, giving an impartial assessment of the disease's physical severity.
- **Prognostic Engine (Tier 4):** A feature fusion layer that combines identified Anatomical Vitality Markers (like ear posture and eye depth) with the visual load. By producing the "Recoverability Probability," this engine advances the system from basic detection to proactive decision support.
- **Explainability (XAI) Reasoner:** To ensure clinical transparency and trust, this module employs a "Chain of Thought" (CoT) attribution framework that enables veterinarians to comprehend the model's internal reasoning.

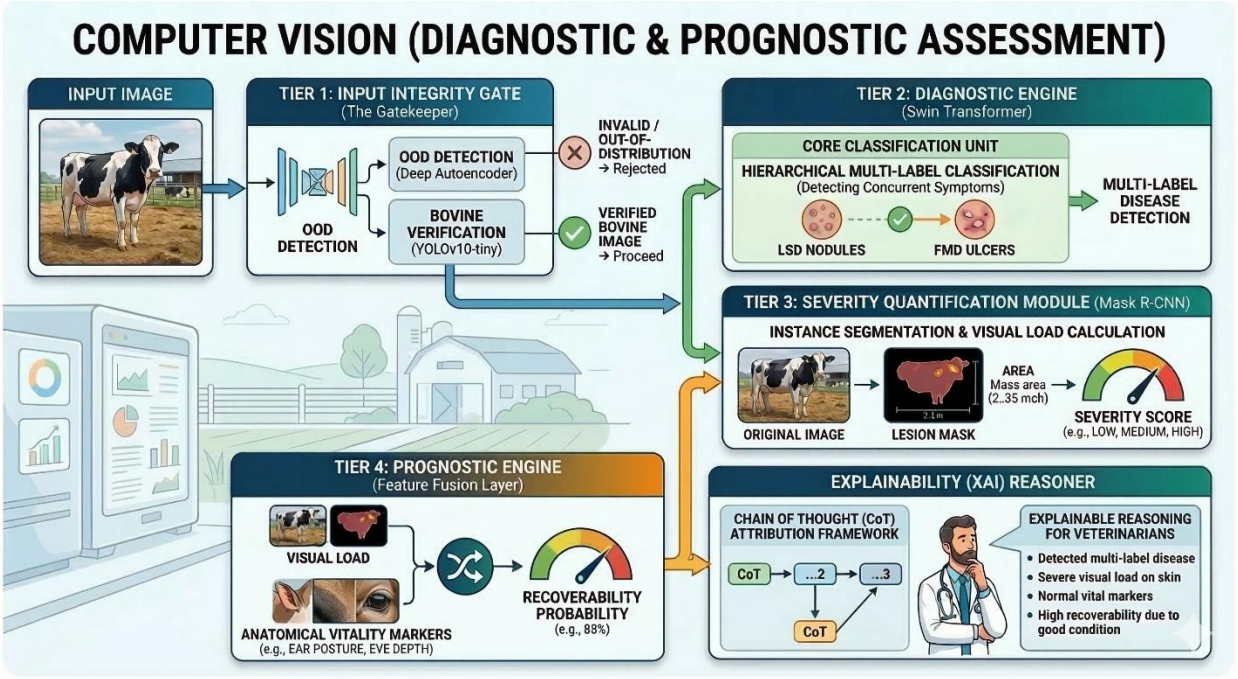


Figure 4 Pipeline of this Component

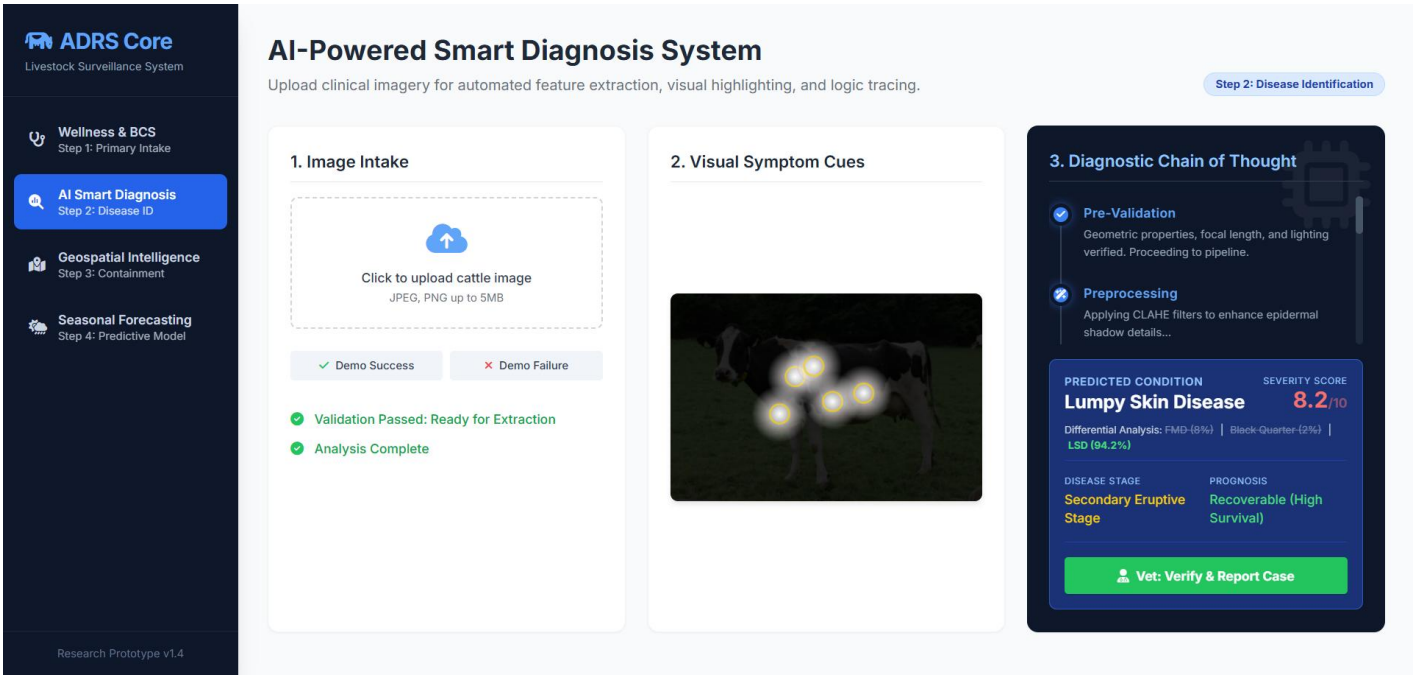


Figure 5 UI Mockup for this Component

4.3 Component Interactions

The CV module keeps in touch with the ADRS ecosystem in both directions:

- **Interaction with Geospatial Module:** Localized risk zones are created when the CV module pushes the disease type and severity score to the Interactive Intelligence Map following a "High-Confidence" diagnosis.
- **Interaction with Wellness Module:** By hiding skin lesions that could otherwise obstruct skeletal keypoint detection, the CV module gives the Wellness Component segmented frames to help with Body Condition Scoring (BCS).
- **Interaction with Seasonal Forecasting Component**: The Seasonal Forecasting component receives the confirmed cases of the disease which had been confirmed by CV module to be used as actual historical data by the forecasting engine.

4.4 Data Flow Explanation

To guarantee computational effectiveness and diagnostic precision, the data flow within the CV component adheres to a rigid linear progression:

1. **Ingestion:** Using the ADRS Mobile App, a 2D image is taken and sent to the cloud via a RESTful API.
2. **Validation:** The Input Gate verifies the existence of the subject and the quality of the image. A rejection signal with a specific reason is sent back to the farmer right away if the image fails (for example, because it is too dark).
3. **Analysis:** Validated images go through the Diagnostic Engine, where the Segmenter covers lesion areas and the Transformer detects disease characteristics.
4. **Prognosis:** The Prognostic Engine correlates these features to determine the disease stage (Early/Mid/Late) and recovery likelihood.
5. **Reporting:** Using WebSockets, the final structured report (Diagnosis + Severity + Prognosis + XAI Heatmap) is pushed to the vet's dashboard and the DAPH surveillance map.

4.5 Scalability and Security Considerations

- **Scalability:** During an outbreak, the CV component is deployed using Docker Containers orchestrated by Kubernetes to manage concurrent uploads from thousands of vets. This enables GPU nodes to scale horizontally during periods of peak reporting.
- **Data Security:** TLS 1.3 encryption is used for all image transmissions. AES-256 is used to encrypt cloud-stored images while they are at rest.
- **Privacy (Anonymization):** To ensure compliance with data privacy ethics, images go through Metadata Stripping to remove GPS coordinates and owner-identifying information before being used for model re-training or shared with the geospatial module.
- **Model Integrity:** The OOD Gate functions as a security layer, rejecting manipulated or digitally altered images that do not adhere to natural livestock photography in order to prevent adversarial attacks (e.g., uploading fake images to trigger false quarantine zones).

5. USER REQUIREMENTS

5.1 Stakeholder Identification

To guarantee a targeted, clinical-grade deployment, the stakeholder group for the Animal Disease Reporting System's Computer Vision (Diagnostic & Prognostic Assessment) component is rigorously defined. Field veterinary officers who work in both public and private settings are the main and only stakeholders. These experts serve as the system architecture's end users and crucial decision-makers. In order to assess field reports, prioritize farm visits during rapid disease outbreaks, and decide on suitable treatment or culling protocols based on the system's prognostic outputs, they use the computer vision component as a remote triage and diagnostic assistant.

5.2 Functional Requirements

The multi-tier computer vision pipeline's unique behaviors and capabilities are determined by the functional requirements, guaranteeing that each stage of the diagnostic procedure is quantifiable, implementable, and in line with the requirements of the veterinary officers. These are enumerated numerically as required by the standard proposal framework:

1. **Image Ingestion and Validation**: The system will take 2D mobile livestock photos and process them right away through an Input Integrity Gate to use object detection to confirm the existence of a bovine subject.

2. **Out-of-Distribution (OOD) Rejection**: Using OOD detection, the system will automatically reject images that are blurry, illuminated incorrectly, or out-of-context (non-animal images). The veterinary officer will receive a specific error message prompting them to submit a clearer image.

3. **Multi-Label Disease Classification**: In order to identify and label several co-occurring skin conditions on a single subject, such as parasitic mange and lumpy skin disease, the system must simultaneously process validated images.

4. **Severity Quantification**: To determine a Bovine Severity Score (BSS), which is the ratio of the infected surface area to the visible healthy skin, the system will segment skin lesions at the pixel level.

5. **Prognostic Scoring**: To produce a Recoverability Probability percentage for the afflicted animal, the system will examine anatomical vitality markers, such as ear posture and eye depth, and combine them with the BSS.

6. **System Integration and Data Handoff**: The system shall automatically transmit the final structured payload, which is composed of diagnosis, severity level, and prognosis, to the central database to trigger the interactive geospatial mapping and wellness modules.

5.3 Non-Functional Requirements

This document specifies the minimum quality, security and performance standards required for the computer vision component to work consistently and reliably when operating in general outdoor service environments.

Performance and Speed

- **Inference Latency**: The user must receive an output as a result of proper processing through all four (4) tiers of the computer vision system in less than five (5) seconds during a standard 4G data service operation.

- **Diagnostic Threshold**: The machine learning classification engine must demonstrate the ability to classify all targeted disease classes with a minimum accuracy and F1-score of eighty-five percent (85%) or greater for the classification to be considered clinically viable.

Security and Privacy

- **Data Encryption**: All image data must be encrypted while in transit between the mobile application and the cloud processing server using Transport Layer Security 1.3 (TLS 1.3) and during rest using Advanced Encryption Standard with a key length of 256 bits (AES-256).

- **Automated Anonymization**: The system will automatically remove any EXIF data, GPS coordinates, and timestamps from the images to ensure the privacy of the farm owner before the images are archived or delivered to other modules.

Usability and Resilience

- **Clinically Readable**: All chain-of-thought explainability reports will be created using universal veterinary terminology; therefore, the user will be able to understand the systems' rationale without a background in machine learning.

- **Intuitive Error Messages**: The Input Gate should give the user information about why there was an error with the submitted input and give clear instructions on how to fix the issue. An example would be, "make sure the subject is positioned correctly and in adequate lighting."

Scalability and Availability

- **Dynamic Cloud Scalability**: The cloud infrastructure must provide the ability to scale the number of GPU container instances to support up to 500 concurrent inferences per minute during disease outbreaks of high density with real-time active data.

- **Offline Graceful Degradation**: The mobile application must store cached images locally in the event of a network outage and afterwards upload them into the inference pipeline as soon as the network is restored.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The ADRS Computer Vision technology will be used directly by field veterinary officers as the end-users, but the market for commercialization will be mainly the agencies and institutions that hire and fund field veterinary officers. The primary market segment will be the Business-to-Government (B2G) market specifically, the Sri Lankan Department of Animal Production and Health (DAPH) and regional provincial councils that are responsible for controlling livestock diseases. The secondary market will consist of Business-to-Business (B2B) customers and include large private dairy cooperatives and commercial veterinarians who require effective herd management solutions.

A veterinary academic as well as a public health specialist, Dr. Thusitha K. Karunarathna (Department of Veterinary Public Health & Pharmacology, University of Peradeniya), is an ideal customer persona for this system.

- Doctor Karunarathna has earned a Ph.D. from the University of London, where he specializes in emerging viral zoonoses, outbreak investigation, and infectious disease epidemiology and leads large-scale surveillance projects (e.g., tracking E. coli in Kandy district cattle) and antimicrobial resistance. The customer persona is highly committed to obtaining scientifically validated, noise-free, in-field data. They often express frustration with conventional reporting techniques due to blurry images and unverified images coming from rural farms distorting epidemiological models, resulting in inaccurate tracking of viral outbreaks. As researchers and teachers creating national public health policies, this persona would be greatly helped by the Out-of-Distribution (OOD) Gating Layer and Chain of Thought (CoT) explainability provided by the system. The system will provide the rigorous, clinical-grade triage necessary to eliminate "junk" data and ensure disease propagation statistics used for academic research and government intervention are accurate.

6.2 Value Proposition

The Value Proposition for the ADRS Computer Vision component can be summed as "Clinical-Grade Triage in the Wild". This Component is reducing Veterinary deployment waste for DAPH and Dairy Cooperatives through it's Out-of-Distribution (OOD) Gating Layer by filtering out information that is either irrelevant or unreadable. This allows for Veterinary Officers' workflow to move from reactive crisis management to proactive data driven prioritised actions. By providing hierarchical multi-label classifications and predictive Bovine Severity Scores on standard 2D mobile devices, we eliminate the requirement of costly 3D sensors; and reduce cases of comorbid conditions being misdiagnosed. It stabilises milk production and decreases livestock mortality by ensuring that the true emergency cases are being seen by a Vet first.

6.3 Revenue Model

The commercialization plan has a 2-part combination of hybrid models of licensing and subscriptions, focusing on institutional purchasers. For the public or government sectors (DAPH), there is an Annual Enterprise Contract as a B2G contractual agreement. This contract allows registered government veterinarians to conduct unlimited inference on the system and will come with a specific SLA, which will provide cloud uptime and semi-annual re-training of the models.

The private sector, represented by dairy cooperatives and private veterinary clinics, will obtain access to the SaaS (Software-as-a-Service) application through a monthly subscription fee, as well as a User License Agreement based on the cooperative's size, so that future scaling for future subscriptions will take place as the cooperatives hire new veterinarians.

6.4 Pricing Strategy

The pricing strategy is designed based on value and penetration strategies, allowing rapid adoption of the product throughout the Sri Lankan agriculture sector. The B2G enterprise license will use a flat, subsidized annual cost, which is consistent with government financial support for agriculture development budgets in order that there are no financial impediments for individual field officers to access and utilize the software. The B2B SaaS pricing structure will follow a tiered pricing schedule based on the number of diagnostic inferences required per month by each farm. Pricing for the software will be set at a dramatically lower price than is currently paid by large farms for traditional hardware-based Precision Livestock Farming (PLF) sensors, creating an irresistible economic incentive for large farms to utilize the software-only ADRS pipeline.

6.5 Competitive Advantage

The Computer Vision component of ADRS has a clear and significant competitive advantage over existing traditional hardware solutions and generic mobile applications for veterinarians. Existing applications are only digital logbooks without automated intelligence, whereas high-end PLF systems are reliant upon costly sensors or 3D cameras that are not economically feasible for local deployment. Additionally, existing models of AI research typically operate in the "closed-world" scenario and thus provide diagnostics that rely on poor data and are unable to diagnose co-morbidities.

The following table demonstrates why the proposed ADRS CV architecture is superior to current market alternatives in addressing the specific needs of Sri Lankan veterinary officers:

Table 1: Competitive Analysis of Livestock Diagnostic Solutions

Feature / Capability	**Proposed ADRS CV Component**	**Standard Vet Mobile Apps**	**Traditional PLF Sensors (3D/Wearables)**	**Generic CNN Research Models**
Hardware Requirement	Standard 2D Smartphone Camera	Standard Smartphone	Expensive 3D Depth Cameras / Collars	High-end GPU processing
Input Integrity (OOD Gating)	Yes (Automated rejection of blurry/junk data)	No (Manual review required)	N/A (Sensor-based)	No (Forces a hallucinated diagnosis)
Comorbidity Detection	Yes (Multi-label Sigmoid classification)	No	No (Focuses on general distress)	No (Single-disease output only)
Prognostic Output	Yes (Bovine Severity Score & Recoverability)	No	No (Tracks only current state)	No (Binary Sick/Healthy output)
Explainability (Trust)	Yes (Chain of Thought logical reporting)	N/A	No (Black-box algorithms)	Rarely (Usually limited to heatmaps)
Implementation Cost	Low (Software-based, scalable)	Low	Extremely High	Moderate to High

6.6 Intellectual Property Considerations

A major component of the continued viability of commercialisation is protecting the CV component's proprietary logic and data. The primary form of IP is the unique architecture of the pipeline that integrates the Diagnostic Gating Layer, Hierarchical Multi-Label Classifier and proprietary logic to compute the Bovine Severity Score (BSS) based on vision methods (Deep Learning in particular). This unique software pipeline and prognostic algorithm is protected as a trade secret with the potential to be patented as appropriate. A critically valuable proprietary data asset is the thoroughly annotated dataset of local tropical cattle diseases via the rigorous evaluation of the domain expert veterinarians. To ensure there will be no dataset leakage to competitors, strict data governance policies (e.g., NDAs for development teams and secure database encryption provisions) will be enforced.

6.7 Cost Estimation

The Computer Vision module in the Centralized System for Real-Time Animal Farm reporting is built on an assumption of lean, cloud-native operational costs. The total cost budget is divided into two major components: a one-time capital expense for develop and training the models and an ongoing operational expense for cloud hosting and maintenance after the models are operational.

To build the initial development environment will require a lot of computing power and will leverage cloud-based GPUs (Amazon EC2-P4 or Google Cloud Compute instances) to train the complex architectures of Swin Transformer and Mask R-CNN over many months. The other major financial cost during the initial training phase will be incurred in collecting primary field data for the development of the model that will consist of the payment of veterinarians who are going into the field to collect primary data by guiding the purchase of video cameras with the goal of creating the “Ground Truth” data sets used to train the model. After the models have been deployed into production, the ongoing operational costs for inferring will be incurred in hosting of API Gateways, performing encrypted cloud storage, and scaling containerized inference nodes during surge outbreaks.

A breakdown of the estimated budget for development, training and first year of operation is shown below in Sri Lanka Rupees (Rs.).

Table 2 Cost Estimation

Component	**Cost**
Travelling Cost (Data Collection)	Rs. 15,000
Cloud Computing (Model Training)	Rs. 15,00/month
Deployment Cost (Inference API)	Rs. 10,000/month
Data Storage (Encrypted Image Database)	Rs. 2,500/month
Total Estimated Budget	Rs. 600,000

7. BUDGET AND JUSTIFICATION

7.1 Justification for Each Expense

Costs for Hardware

Cloud environments are to do most of the heavy lifting for deep learning training, and therefore, expensive GPU workstations for local use are not necessary. However, an external 1TB Portable Solid-State Drive (SSD) will be required at a cost of Rs. 20,000. This hardware will allow us to securely store, backup and transport the large, high-resolution bovine image datasets that are collected in the field prior to uploading them to the cloud for processing.

Costs for Software Licenses

To train complex architectures such as Swin Transformers, YOLOv10 and Mask R-CNN, we must access a significant amount of computational power. It is sensible to purchase a 6-month subscription to Google Colab Pro to gain access to premium cloud-based GPUs, like the NVIDIA T4 or V100, which will provide faster iterations of training while also ensuring that we do not experience session timeouts (which can happen on free tiers). The total estimated cost of Rs. 3,200/month for 6 months is therefore Rs. 19,200, and is essential to meeting project deadlines. Additionally, open-source tools such as Roboflow will be used for annotating images at the free-tier academic level, so there will be no additional software costs.

Cloud Solutions

To emulate an actual field solution that is very reliable and always on, the models that were trained can be sent to the cloud so that a reliable infrastructure can back them up and provide a basis for an in-production environment. The Inference API service will incur a monthly cost of Rs. 6073 with a total cost during the pilot phase of Rs. 36438. This will provide support for processing incoming request (by mobile devices) with as low a response time as possible. Additionally, an appropriate attribution to the data will be used for data storage of the encrypted and anonymized images in the cloud at Rs. 2,500 per month, for a total cost of Rs. 15,000, in order to facilitate compliance with the respective image databases that generate the OOD checks and diagnostic history.

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) for the Animal Disease Reporting System (ADRS) was created to provide a way to evenly distribute work among the research team and ensure that they are developing each component of the ADRS at the same time so that everything has the same timeline. The project is broken out into 4 components of research assigned to 1 researcher per component. The research and development of the ADRS will take place over a single academic year or 9-months, from the initial data collection to the completed integration of the system.

8.1 Task Breakdown by member

Component 1: AI-Powered Smart Diagnosis (Computer Vision and Prognostics)

- Member 1: A.L.M Athulathmudali
- Key Tasks:

1. Collect and Annotate Primary and Secondary Data Sets for Local Cattle Breeds (i.e. Cattle Breeds found locally) for multiple concurrent diseases and their associated vitality indicators.
2. Design and Build Input Integrity Gate using YOLOv10-tiny and Out-of-Distribution (OOD) Additional Information.
3. Train Hierarchical Multi-Label Classifier (Swin Transformer) and Severity Quantification Module (Mask R-CNN).
4. Implement the Chain of Thought (CoT) transparent framework and prognostic survival engine.
5. Validate the diagnostic threshold using K-Fold cross-validation to ensure it has an 85% accuracy rate.

Component 2: Real-Time Geospatial Clustering & Trajectory Mapping

- Member 2: G.A Sandaru
- Key Tasks:

1. Gather and process both geographical and historical outbreak data regarding Sri Lanka's livestock.
2. Create density-based spatial clustering algorithms to detect contemporary (real-time) disease hot-spots.
3. Utilize physics-based spread modeling to determine both the localized outbreak velocities and trajectories.
4. Build an Interactive Intelligence Map for use by Government and Veterinary Agencies.
5. Build API endpoints to securely accept diagnostic payloads (type of disease and its severity) from the Computer Vision component.

Component 3: Seasonal Forecasting for Risk assessment

- Member 3: S.S Kumarasinghe
- Key Tasks:

1. Combine historical climate records, monsoon characteristics, and environmental variables related to tropical livestock diseases into a comprehensive data set.
2. Design non-linear machine learning forecast models (LSTM's or Conformal Inference) to enable effective temporal forecasting and the identification of sudden spikes in disease occurrence.
3. Create a local, seasonal "Risk Index" by combining relevant weather variables to designate the level of risk within select geographic areas.
4. Confirm the validity of the forecast model's precision when correlating the forecasted outbreak occurrences with recorded outbreak data.
5. Integrate the results of the predictive risk assessment with the central ADRS database to notify vulnerable farming areas.

Component 4: Animal Health Detection (Wellness and Management)

- Member 4: W.M.P.J Wijenayake
- Key Tasks:

1. Develop and establish corporate standard Multiple Base Parameters for the recognition of physiological distress and physical wellness in local dairy cattle.
2. Create a 2D vision-based algorithm to automate the reading of Body Condition Scores (BCS) by using skeletal keypoint detection software.
3. Use the clean segmented masks of bovine animals created using algorithms from the Computer Vision Component to remove lesions when calculating the BCS.
4. Create a farmer view of the wellness of the animal that includes recording the physiological conditions and energy status of the herd.
5. Conduct field tests to compare automated reading of BCS to manually recorded readings prepared by veterinarians.

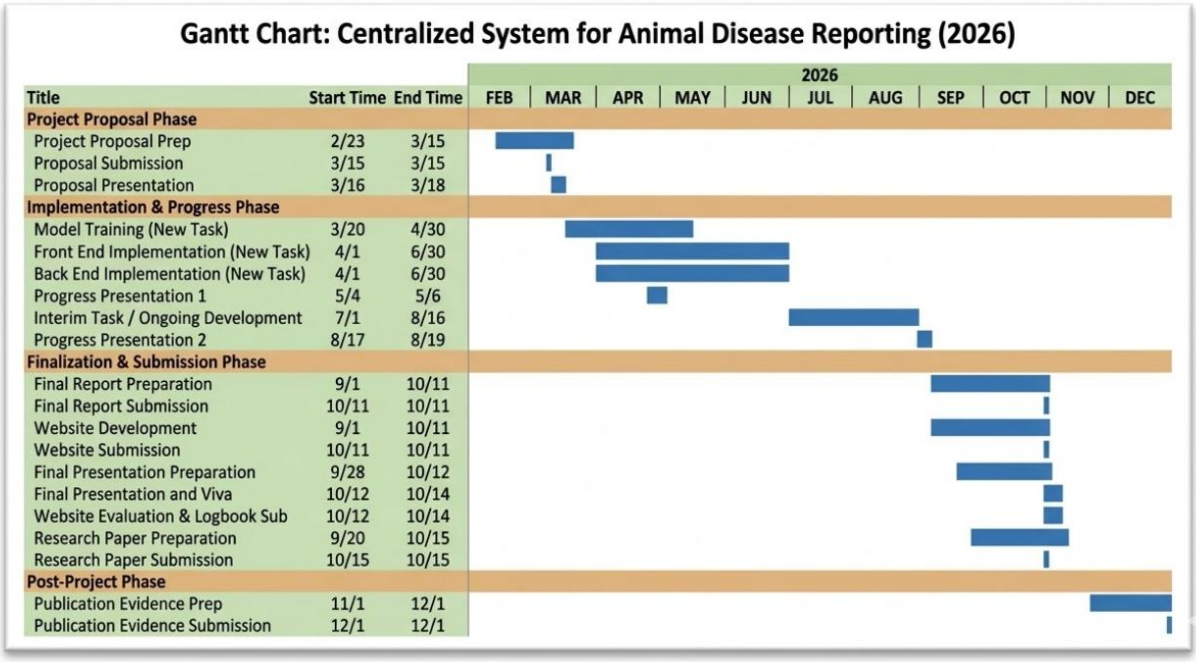
8.2 Timeline and Workload Distribution

To ensure the workload is distributed evenly, each of the 4 Research Team members will complete the core software engineering phases simultaneously. The concurrent development methodology will help avoid any integration bottlenecks, as well as ensure that all module dependencies are available before the scheduled due date.

Table 3 Timeline Alignment and Phase Distribution

Project Phase	**Timeline**	**Primary Activities (All Members)**	**Milestones / Deliverables**
Phase 1: Requirements & Data Acquisition	Months 1 - 2	Literature review, defining system architecture, ethical clearances, field data collection, and dataset annotation.	Finalized Proposal & Initial Datasets Completed.
Phase 2: Core Algorithm Development	Months 3 - 5	Individual model training (Vision, Geospatial, Forecasting, BCS), hyperparameter tuning, and local module testing.	Core Models Trained & Baseline Accuracy Achieved.
Phase 3: System Integration & XAI	Months 6 - 7	Establishing cloud API connections, database synchronization, implementing explainability (CoT) features.	Integrated ADRS Prototype Ready.
Phase 4: Field Testing & Validation	Month 8	System deployment on standard devices, live testing with veterinary officers, performance metric evaluation.	Validation Report & System Refinements.
Phase 5: Documentation & Finalization	Month 9	Finalizing the research thesis, formatting codebase, publishing user manuals, and presentation prep.	Final Research Document & System Handover.

9. GRANTT CHART



Title	Start Time	End Time
Project Proposal Phase		
Project Proposal Prep	2/23	3/15
Proposal Submission	3/15	3/15
Proposal Presentation	3/16	3/18
Implementation & Progress Phase		
Model Training (New Task)	3/20	4/30
Front End Implementation (New Task)	4/1	6/30
Back End Implementation (New Task)	4/1	6/30
Progress Presentation 1	5/4	5/6
Interim Task / Ongoing Development	7/1	8/16
Progress Presentation 2	8/17	8/19
Finalization & Submission Phase		
Final Report Preparation	9/1	10/11
Final Report Submission	10/11	10/11
Website Development	9/1	10/11
Website Submission	10/11	10/11
Final Presentation Preparation	9/28	10/12
Final Presentation and Viva	10/12	10/14
Website Evaluation & Logbook Sub	10/12	10/14
Research Paper Preparation	9/20	10/15
Research Paper Submission	10/15	10/15
Post-Project Phase		
Publication Evidence Prep	11/1	12/1
Publication Evidence Submission	12/1	12/1

Figure 6 Grantt Chart

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